Janelle N. A. Yankey

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Education

Howard University Washington, D.C.

Bachelor of Science in Computer Science, GPA: 3.62 *Jan. 2024 – May 2027*

- Dean's List (Spring 2024, Spring 2025, Fall 2025, Spring 2026)
- **Relevant Coursework:** Fundamentals of Algorithms, Large Scale Programming, Operating Systems, Computer Organization, Computer Networks, Applied Data Science

Experience

Undergraduate Research Intern, Elevate Black Voices × Google Apr. 2026 – Present

Howard University – HCAI *Washington, D.C.*

- Developed an interactive data visualization app using React, Vite, and globe.gl, featuring dot clustering and emotion heatmap overlays, to map AAVE speech pattern data from 294 respondents across 16 U.S. states, surfacing disparities in voice assistant technology performance.
- Built a Python NLP sentiment analysis pipeline using a fine-tuned RoBERTa transformer model to classify 15 IRB-approved respondent audio samples (a subset of 294 total), configured for fully offline execution to maintain IRB data privacy compliance.
- Co-authored a section of an academic paper on HCI bias affecting AAVE speakers, analyzing how voice assistant design perpetuates disparities, currently in preparation for publication.

Research Intern, Office of Naval Research (ONR) Apr. 2025 – Mar. 2026

Howard University – HCAI *Washington, D.C.*

- Analyzed and annotated 198 LLM-generated threat-assessment responses across RAG and non-RAG conditions, evaluating comprehension, reasoning quality, and failure patterns to inform model improvements for Howard HCAI's ONR-funded TADMUS research.
- Evaluated AI-generated image augmentation quality across a 98-image U.S. Navy vessel dataset using Adobe Generative Fill, applying a five-criteria, color-coded rating system validated with inter-rater reliability measures.
- Conducted HCI usability analysis of user trust and mental model formation in human-AI interaction, informing design improvements for Howard HCAI's RAG-based decision support tool.

Publications

Adinkra Symbols through Generative AI: A Semiotic Analysis of Emoji-Based Translation Jun. 2026

- J. Nias, S. K. Aryal, K. Campbell, K. Cooper, T. Douglass, **J. N. A. Yankey**. *ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, pp. 1340–1354.

CARE: Culturally Aligned AI for Reflective Creativity Through Adinkra Semiotic Knowledge May 2026

- J. Nias, K. Campbell, **J. Yankey**, T. Douglass. *Proceedings of the AAAI Symposium Series*, 8(1), pp. 706–714.

Projects

FinReq – Financial Request System | *TanStack Start, Supabase, PostgreSQL, Power Automate* Aug. 2026 – Present

- Built 5 merged features for a team-built university financial request system (Senior Project): a rules-based approval routing engine, a Power Automate workflow trigger, three-layer data validation, trigger-based audit logging, and automatic record linking in PostgreSQL with row level security.

Weather Outfit Picker | *React, Vite, Express.js, Claude API* Jun. 2026 – Present

- Developing an AI-powered outfit recommendation app using the Open-Meteo API and Claude to generate weather- and style-aware suggestions, with a secure Express.js backend keeping the API key server-side.
- Engineered a Scan My Outfit feature using Claude's vision API to analyze photographed outfits, returning a 0–100 weather-compatibility score and styling suggestions.

Let's Prep – Study & Quiz App | *Python, SQLite, Full-Stack Development* Sep. 2025 – Dec. 2025

- Co-developed a full-stack study app (3-person team) that parses PDFs, Word docs, PowerPoints, and images to auto-generate flashcards, MCQs, and fill-in-the-blank quizzes, with authentication, quiz history, and PDF export.

MedTrack – Health Monitoring App | *Java, JavaScript, MySQL, PHP, Python* Mar. 2025

- Built a full-stack health monitoring app in 24 hours at BisonBytes Hackathon 2025, featuring real-time condition tracking, automated reminders, and provider updates.

Certifications & Professional Development

AWS Certified Cloud Practitioner | *Amazon Web Services* Apr. 2026 – Apr. 2029

CodePath | *AI Engineering, Cybersecurity, Technical Interview Prep* Jun. 2026 – Aug. 2026

- Completed three concurrent 10-week industry-led courses in AI engineering, cybersecurity, and technical interview preparation, including hands-on labs in steganography, password cracking, penetration testing, and digital forensics.
- Built VibeMatch, a music recommender system with a custom guardrail and reliability layer, as the AI Engineering (AI110) capstone project.

Leadership & Extracurricular

ColorStack | *Member* 2026 – Present

Rewriting the Code | *Member* Jun. 2026 – Present

Technical Skills

Languages: Python, Java, JavaScript, SQL, HTML/CSS, C++, PHP

Frameworks & Libraries: React, Vite, Express.js, globe.gl, RoBERTa/NLP, JavaFX, JUnit

Developer Tools: VS Code, Git/GitHub, Android Studio, Google Cloud Platform, AWS

Concepts: Object-Oriented Programming, REST APIs, Relational Databases, Data Visualization, Full-Stack Development